Statement on Supporting Georgia Tech's Mission and Values

Georgia Tech's mission — to develop leaders who advance technology and improve the human condition — is not an abstraction to me; it describes, almost exactly, the arc of a career that has moved between materials science, clinical medicine, neuropsychiatric research, and large-scale industrial operations. I am a proud Mexican-American scientist and engineer, bilingual in English and Spanish, and I believe the value of that path lies in its unconventional shape, not despite it: I bring the discipline of someone who has had to make rigor and quality tangible in a hospital lab, at a lab bench, and on an industrial floor, not just in a manuscript.

Students First

I have spent more of my career teaching and developing people than I expected to, and it is the part of the work I value most. As a doctoral researcher, I mentored more than thirty undergraduate and graduate students in electron microscopy and 3D bioprinting and taught the microscopy module of a biomaterials characterization course — work that produced a co-authored publication, with students as collaborators, not assistants. Before that, I taught high school physics, chemistry, and biology at every level from remedial to AP, and spent years tutoring students, often ones who could not otherwise afford it. At Amazon, I built a training pipeline for Process Assistants where none had existed, trained fifteen of them, and was later asked by my own manager to train the Operations Manager and several Area Managers. Whether the student is nineteen and pipetting for the first time or forty and running a warehouse floor, the job is the same: build the scaffolding that lets someone do what they could not do before, then get out of the way.

Excellence and Innovation

Georgia Tech's insistence on excellence and its celebration of innovation are, to me, two names for the same habit: refusing "good enough" when better is measurable. My doctoral work on a conductive carbon-nanotube–alginate hydrogel required characterizing colloidal stability across eleven surfactant systems and measuring bulk conductivity until I could say precisely which formulation worked and why. At Amazon, that same instinct produced an 82% improvement in dock throughput and cut process recirculation from 5% to under 2%, by building the data systems needed to see a problem clearly before fixing it. Since leaving full-time academia, I have kept building — independently, without a lab or funding — a biomanufacturing quality-intelligence platform, an image-to-bioprint-scaffold pipeline, and computational studies of psychiatric drug targets and glial biomarkers, because I could not let the questions go unanswered. That is the freedom to inquire Georgia Tech says it safeguards; I have been practicing it on my own time for years.

Collaboration Across Difference

Nearly everything I have built has depended on people whose training looked nothing like mine. My doctoral research existed because a materials lab at UTEP collaborated with a bioengineering lab at Rice University. My postdoctoral work depended on partnerships with a regional brain bank and a county coroner's office — institutions with very different standards for what counts as care with someone's tissue. My seven peer-reviewed publications were each the work of four-to-twelve-author teams spanning chemistry, bioengineering, and medicine. And at Amazon, the discipline I lead requires daily

coordination across process owners, associates, and peer managers with no shared academic background at all. Collaboration across genuine difference is not a soft skill adjacent to the research — it is usually the actual bottleneck, and I have spent my career getting good at unblocking it.

Acting Ethically, With the Receipts to Prove It

The clearest way I can demonstrate ethical practice is to point to what I have chosen to publish about my own work when no one required it. In an independent computational pharmacology study, I reported plainly that my model did not reproduce the experimentally observed effect I was testing for, and discussed why. In a biomarker analysis of a psychiatric patient cohort, I self-graded my own work, documented eight methodological errors in my own prior analysis, and corrected them in the open. This did not start after my PhD — it is the same discipline CLIA and COLA compliance demanded of me as a clinical laboratory supervisor overseeing more than a thousand patient toxicology and pharmacogenomic samples a month, where an unreported error is a patient-safety failure, not an academic embarrassment. I hold my own research to that standard because I no longer know how to hold it to a lower one.

Stewardship and Community

Responsible stewardship, to me, includes the unglamorous work of chain-of-custody documentation for postmortem brain tissue, quality-compliance systems holding at 99.7%, and regulatory risk documentation for research-use-only devices — work that rarely appears in a results section but is the reason the results can be trusted. It also includes community work I kept doing regardless of what else was on my plate: co-founding a fundraising and mentoring effort for pre-medical students preparing for the MCAT, serving as a science fair judge at a local high school, and volunteering with a food pantry restoration effort through my engineering honor society. None of it was required.

Why This Matters at Georgia Tech

I am not applying to Georgia Tech because it is convenient to my background. I am applying because the combination of rigor, translational ambition, and institutional scale the Coulter Department represents is the only environment where I can picture using everything I have built a career learning — materials science, clinical regulatory discipline, neuropsychiatric research, and operational scale-up expertise most faculty candidates never have reason to acquire. I would bring that combination, and the values it was built on, to the students and colleagues I would be lucky to work alongside here.